

National Institute of Technology Hamirpur
Department of Computer Science and Engineering
CS-414 Information Security and Privacy Lab
Assignment- 3

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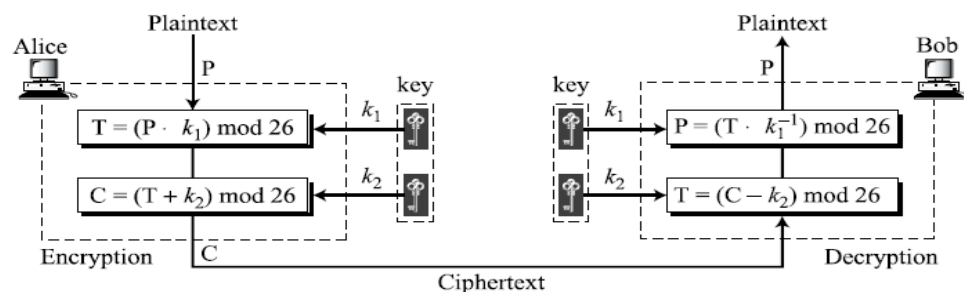
1. Write a program to for Affine Cipher showcasing the encryption as well as decryption algorithm for the given plaintext and ciphertext:

Plaintext for Encryption: “hello”

Ciphertext for Decryption: “ZEBBW”

Key: (7,2)

Figure 3.11 *Affine cipher*



In the affine cipher, the relationship between the plaintext P and the ciphertext C is

$$C = (P \times k_1 + k_2) \bmod 26$$

$$P = ((C - k_2) \times k_1^{-1}) \bmod 26$$

where k_1^{-1} is the multiplicative inverse of k_1 and $-k_2$ is the additive inverse of k_2